

HARIPRIYA M A

C++ Software Engineer | Python Developer | Backend & Systems Development

Dubai, UAE | UAE Residence Visa | Immediately Open to UAE-Based & Remote (Europe) Roles
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GitHub: github.com/CodeWithHaripriya | Portfolio: codewithharipriya.github.io/haripriya-portfolio

PROFESSIONAL SUMMARY

C++ and Python Software Engineer with 4+ years of experience building performance-critical backend systems and full-stack web applications. Background includes developing and testing C++ medical imaging software for clinical diagnostic hardware (multithreading, memory management, static/dynamic analysis with Coverity and Valgrind), and full-stack development in Python/Django (SQL, REST APIs, responsive UI). Comfortable owning the full Software Development Life Cycle (SDLC) - from requirements through production support - in Agile teams. Currently based in Dubai (UAE Residence Visa) and immediately available for C++ Developer or Python Developer roles - UAE-based, hybrid, or remote with European teams.

SKILLS

Programming Languages: C++ (Linux & Windows), Python, Java (Working Knowledge), JavaScript/TypeScript (Working Exposure)

Python & Web Development: Django, HTML5, CSS3, Bootstrap, Tkinter (GUI), REST API Development, Python Automation

C++ Engineering: Object-Oriented Programming (OOP), Design Patterns, Multithreading & Concurrency, Memory Management, STL, Data Structures & Algorithms (DSA), Boost.Asio, TCP/IP & Socket Programming, Lock-Free Programming Concepts, Unit Testing (GTest)

Backend & Architecture: RESTful APIs, Microservices, gRPC, Message Queues (Kafka), CMake

Databases: PostgreSQL, MongoDB, SQL

Debugging & Quality: Coverity Static Code Analysis, Valgrind, Root Cause Analysis, Code Reviews, Defect Reduction

Tools & Platforms: Docker, Git, GitHub, SVN, JIRA, Redmine, CI/CD Pipelines, Linux, Windows

Practices: Agile/Scrum, Waterfall, Technical Documentation, SDLC Ownership, Cross-Functional Collaboration, Mentoring

WORK EXPERIENCE

Software Engineer

September 2025 – February 2026

Trenser Technology Solutions Pvt Ltd — Trivandrum, India

- Engineered high-performance, maintainable applications using object-oriented design principles as part of a team, contributing to a modular codebase used by enterprise medical imaging clients.
- Gained hands-on exposure to JavaScript, TypeScript, and React.js, building UI modules for the same enterprise medical imaging platform, strengthening cross-stack collaboration with frontend teams.

Software Engineer

June 2022 – September 2025

Quest Global Engineering Services Pvt Ltd — Trivandrum, India

- Developed and tested features for Canon Medical Systems MRI scanning software in C++, supporting critical diagnostic hardware used in clinical environments globally.
- Owned the full SDLC - requirements analysis, technical design, development, testing, deployment, and production support.
- Reduced application defect rates by approximately 20-25% through systematic debugging, root cause analysis, and Coverity static code analysis.
- Built Python automation and testing utilities that reduced manual regression verification effort.
- Architected a C++ producer-consumer streaming framework (StreamHub) for high-throughput binary data transfer, using async chunk transfer and memory pooling to eliminate malloc/free overhead via a pre-allocated buffer pool.
- Debugged complex memory ownership issues using Valgrind, ensuring production-level stability.
- Received the "On The Fly" Award twice (May 2023 & Dec 2024) for high-quality delivery on critical international project modules.

Python Developer

August 2021 – February 2022

Infox Technologies — Kochi, India

- Developed and maintained full-stack web applications using Python, Django, HTML5, CSS3, and Bootstrap, ensuring responsive user interfaces and robust backend logic.

- Designed and executed relational database schemas and queries using SQL to support efficient data management and application functionality.
- Built desktop GUI applications utilizing Python's Tkinter library for customized internal utilities and tools.
- Onboarded and mentored incoming developers and trainees on core Python programming, Django framework fundamentals, and web development best practices.
- Collaborated in an agile team environment to troubleshoot, debug, and optimize application code, enhancing performance and user experience.

PROJECTS

LuxeScents (E-Commerce Web App): Full-stack e-commerce application built with React, Node.js, Express, and MongoDB, featuring product catalog, cart, and responsive UI. Live: e-commerce-led4.vercel.app | GitHub: github.com/CodeWithHaripriya/E-Commerce

AsyncNet-Server (C++ Network Server): Multi-threaded asynchronous TCP server built with C++17/20 and Boost.Asio. GitHub: github.com/CodeWithHaripriya/ObjectOrientedProgramming/tree/main/01_AsyncNet_Server

LOB-Engine (Order Book Matching Engine): Limit order book matching engine built with C++20, exploring order-matching logic used in trading systems. GitHub: github.com/CodeWithHaripriya/ObjectOrientedProgramming/tree/main/02_LOB_Engine

EDUCATION

Bachelor of Engineering in Electronics and Communication

Ilahia School of Science and Technology, APJ Abdul Kalam Technological University | 2016 - 2020 | Kerala, India

LANGUAGES

English, Tamil, Malayalam, Hindi